Hussein Sharadga

hssharadga@tamu.edu | (607) 759–8526 | Austin, Texas | Website

Education

Texas A&M University, College Station, Texas | 2019 – 2022
Ph.D. in Mechanical Engineering

Jordan University of Science & Technology, Irbid, Jordan | 2015 – 2017
M.Sc. in Mechanical Engineering

Al-Balqa' Applied University, Irbid, Jordan | 2011 – 2015
B.Sc. in Mechanical Engineering

Academic Positions

Assistant Professor | 2026 – Present
Prairie View A&M University, Mechanical Engineering

Research Affiliate | 2024 – Present
University of Texas at Austin

Assistant Professor | 2024 – 2026
School of Engineering, Texas A&M University- International

Postdoc Researcher | 2022 – 2024
University of Texas at Austin

Research Focus

AI & **GPU-Accelerated Optimization**
for Large-Scale Power Grids

Awards

ARPA-E Grid Optimization Competition- Challenge 3 Prize Winner | 2023
University of Texas at Austin
Trial 1: ranked 2nd in the total score
Trial 2: **ranked 1st**
Trial 3: ranked 2nd **|$115,000 prize|**

Publications

Google Scholar: **h-index 9; 832 citations** (Sep. 2026)

1) **H. Sharadga**, Javad Mohammadi, "A GPU-based Solver for Scalable Unit Commitment in Large Power Grids", *IEEE Transactions on Industry Applications*, 2026.

2) **H. Sharadga**, Y Du, J Mohammadi, "Rapid Concurrent GPU-CPU Solvers for Scalable Unit Commitment in Large Power Grids", *IEEE SmartGridComm*, 2026.

3) **H. Sharadga**, Yuhan Du, Javad Mohammadi, "Probabilistic-Adversarial Networks (PANs): A Probability-Informed Deep Learning Approach for Electrical Demand Forecasting", *IEEE Journal of Power and Energy*, 2026.

4) **H. Sharadga**, Javad Mohammadi, "GPU-Accelerated Optimization Solver for Unit Commitment in Large-Scale Power Grids", *IEEE Texas Power and Energy Conference*, 2026

5) **H. Sharadga**, Abdullah Hayajneh, Erchin Serpedin, "Evaluating AI-based Image Inpainting Techniques for Facial Components Restoration Using Semantic Masks", *AI Journal*, 2026.

6) A Dawahdeh, **H. Sharadga**, G Zakeri, A Hayajneh, G Pritchard, "Real-time building energy storage scheduling under electrical load uncertainty: A Markov decision process approach with comprehensive analysis of different pricing policies", *Energy Reports*, 2026.

7) **H. Sharadga**, Javad Mohammadi, "Scalable Unit Commitment for Large Power Grids: Relaxation, Tightening, and GPU-Based Solvers Evaluation", *IEEE North American Power Symposium*, 2025.

8) Maureen S Golan, **H. Sharadga**, Javad Mohammadi, "Power grid resilience: insights from sensitivity analysis using a comprehensive AC optimal power flow model", *Environment Systems and Decisions Journal*, 2025.

9) **H. Sharadga**, Javad Mohammadi, Constance Crozier, Kyri Baker, "Scalable Solutions for Security-Constrained Optimal Power Flow with Multiple Time Steps", *IEEE Transactions on Industry Applications*, 2025.

10) **H. Sharadga**, Javad Mohammadi, Constance Crozier, Kyri Baker, "Optimizing Multi-Timestep Security-Constrained Optimal Power Flow for Large Power Grids", *IEEE Texas Power and Energy Conference*, 2024.

11) **H. Sharadga**, Ahmad Dawahdeh, "Novel MPPT Controller Augmented with Neural Network for Use with Photovoltaic Systems Experiencing Rapid Solar Radiation Changes", *Sustainability Journal*, 2024.

12) Shima Hajimirza, **H. Sharadga**, "Learning Thermal Radiative Properties of Porous Media from Engineered Geometric Features", *Int. J. Heat Mass Transf*, 2021.

13) **H. Sharadga**, Shima Hajimirza, Robert S Balog, "Time Series Forecasting of Solar Power Generation for Large-Scale Photovoltaic Plants", *Renewable Energy*, 2020.

14) Ali Akbar Shafi, **H. Sharadga**, Shima Hajimirza, "Design of Optimal Power Point Tracking Controller Using Forecasted Photovoltaic Power and Demand", *IEEE Transactions on Sustainable Energy*, 2019.

15) **H. Sharadga**, A Dawahdeh, MA Al-Nimr, "A hybrid PV/T and Kalina cycle for power generation", *International Journal of Energy Research*, 2018.

16) M Al-Nimr, S Kiwan, **H. Sharadga**, "A hybrid TEG/wind system using concentrated solar energy and chimney effect", *International Journal of Energy Research*, 2018.

17) MA Al-Nimr, S Kiwan, **H. Sharadga**, "Simulation of a novel hybrid solar photovoltaic/wind system to maintain the cell surface temperature and to generate electricity", *International Journal of Energy Research*, 2018.

Teaching Experience

Prairie View A&M University | 2026 – Present
Assistant Professor

- Dynamics, Computer-Aided Design, and Measurement & Instrumentation Lab.

Texas A&M International University | 2024 – 2026
Assistant Professor

- Statistics, Eng. Modeling & Design, CAD, and Smart Grid Optimization.
- **Student Evaluations: 4.75/5** (Average)

University of Texas Permian Basin | 2023 – 2024
Visiting Assistant Professor

- Thermodynamics, Fluid Mechanics, Heat Transfer, Engineering Graphics, Computer-Aided Mechanical Design, Dynamics, and Thermo-fluid & Mech Sys Lab.
- **Student Evaluations: 4.23/5** (Average) **[Evaluation Report]**

Professional Activities & Service

1. Invited Talks

ARPA-E Grid Software Annual Review, Electric-Stampede's Approach: Fast and Robust Strategies for Large-scale mixed-integer SCOPF.

2. Mentoring

1. Karol Gonzalez, B.Sc. Student, 2026, Texas A&M International University.
2. Julio Martinez, B.Sc. Student, 2026, Texas A&M International University.
3. Maureen S. Golan, Ph.D. Student, 2023, The University of Texas at Austin.

3. Poster Presentations

1. Electric Stampede: A Robust Solution to The Challenging Task of Us Power Grid Optimization, INFORMS Annual Meeting, Phoenix (2023).
2. Time Series Forecasting of Solar Power Generation for Large-Scale Photovoltaic Plants, Texas A&M Conference on Energy (2019).
3. A Fast and Accurate Single Diode Model for Photovoltaic Design, IMECE, Utah (2019).

4. University Service

Search Committee, TAMIU (2024); ABET Committee, TAMIU (2024); ABET Committee, UTPB (2023).

5. Conference Service

Session Chair, IEEE TPEC 2026; Session Chair, NAPS 2025.

Programming Skills

Languages: Python, MATLAB, Julia, C/C++.

AI/ML: Keras, PyTorch, Scikit-learn.

Optimization: Gurobi, MOSEK, IPOPT, COPT, HiGHS.

HPC: **GPU**-accelerated computing for large-scale optimization; parallel computing.